MILITARY SPECIFICATION SHEET

STARTER, AIR TURBINE, ACCESSORY DRIVE COMBINATION, AIRCRAFT ENGINE, MODEL ATS/AD-1

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the starter described herein shall consist of this document and the latest issue of Specification MIL-S-22518(AS).

REQUIREMENTS

Applicable paragraphs of MIL-S-22518A(AS)		
3.8.1	Envelope:	Figure 1
3.7	Wt, Max (Lb):	32
4.5.19	Aircraft Model:	F-8
4.5.20	Engine Model:	J57
3.8.2.2	Accessory Drive:	
	Engine:	Figure 1
	Pad "B":	Figure 1
3.5.5 3.9.1	Rotation Viewed from Pad "B" End:	
	Pad "A":	Clockwise
	Pad "B":	Counterclockwise
3.3.2	Ambient Temperature Operating Range (°F):	
	Starter:	-65 to +160
	Accessory Drive:	-65 to +250
3.3.2	Exposure Temperature Range (°F):	-100 to +275
3.3.2.2 3.5.7 4.5.1.1 4.5.7	Control Valve:	
	Opening Rate, Max:	22 psig per second pressure rate for first 0.5 second; full open in 3 to 5 seconds
	Closing Rate, Max:	0.5 second

FSC 2995

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REQUIREMENTS (Continued)

Applicable paragraphs
of MIL-S-22518A(AS)

3.5.1 Rated Conditions:

Starter (Pad "A"):
Drive Speed. RPM: 1200
Drive Torque, Min (Lb-Ft): 290
Airflow, Max (Lb per Min): 110
Inlet Air Total Pressure (In. Hg A): 100
Inlet Air Total Temperature (°F): 345
Exhaust Gas Discharge Pressure (In. Hg A): 29.9

Accessory Drive (Pad "B"):
Continuous: 10 HP at 4200 RPM
Intermittent: 32 HP at 4200 RPM for 5 minutes
Maximum: 55 HP at 7000 RPM Pad "A" speed for period specified in 4.5.7

NOTE: At an input speed of 7000 RPM at pad "A" drive, pad "B" drive shall rotate at 4200 +10 RPM.

3.5.1 cutout RPM: 3200 plus 200 minus 0

3.5.2
3.5.8 Maximum Temperature Limit: The air turbine shall be capable of operating with a maximum inlet air total temperature limit of 650 F at rated inlet air total pressure.

3.5.8
4.5.17 Altitude: The ATS/AD shall not be damaged when subjected to 75,000 ft altitude in the overrunning mode of operation. The ATS/AD shall be capable of developing rated drive speed and torque at an 8000 ft altitude in the starter mode of operation.

Accessory Drive: Complete 20 cycles of the test schedule of Figure 2.

3.5.9
4.5.16 Attitude: The ATS/AD shall be capable of meeting the operational requirements when the ATS/AD centerline, taken through the pad "A" shaft axis, is in any position within 10 degrees of the horizontal position. Inverted flight test time is 15 minutes.

3.5.10
4.5.10 Overrunning (Qualification): The ATS/AD shall be overrun at a 7250 RPM pad "A" drive speed for a total of 20 hours. Testing need not be continuous but test periods shall not be less than 5 hours.

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REQUIREMENTS (Continued)

Applicable paragraphs
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3.5.10 Overrunning (Acceptance): The ATS/AD shall be overrun
4.5.10 at a pad "A" speed of 7250 RPM and subjected to a pad "B" load of 10 HP for 5 minutes.

4.5.4 Initial Calibration:

Pad "A" Drive

Torque (Lb-Ft)	Speed RPM	Speed Condition
370 Min, 465 Max	0	Stall
290 Min ---	1200	Rated
50 Min ---	3200 to 3400	cutout

Airflow, Max (Lb per Min)	Speed RPM	Speed Condition
110	0	Stall

4.5.5 No-Load Operation (Qualification):
Time (Second): 60
Pad "A" Drive Speed (RPM), Min: 3600

4.5.5.1 Burst Test:
Inlet Air Total Pressure (PSIA): 75
Inlet Air Total Temperature (°F): 445 ±10°F

4.5.6 No-Load Operation (Acceptance): Pad "A" Drive Speed (RPM), Min: 3600

3.5.5 Endurance Test:
4.5.7 Accessory Drive: The ATS/AD shall be driven for 960 hours at 7000 RPM with a continuous load of 10 HP applied at pad "B." A load of 32 HP shall. be applied at pad "B" for 5 minutes of every hour during the above 960-hour endurance test. This phase of endurance testing need not be continuous; however, test periods shall not be less than 5 hours duration. This endurance testing may be combined with the tests for the starter in this paragraph in an attempt to simulate service conditions.

Accessory Drive High Torque Test: At a pad "A" drive speed of 7000 RPM the ATS/AD shall be cycles for 15 minutes at 10 HP and 15 minutes at 55 HP for a total of 2 hours.

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REQUIREMENTS (Continued)

Applicable paragraphs
of MIL-S-22518A(AS)

Starter: No. of cycles: 1200. Each cycle shall consist of three consecutive phases as follows:

Test Stand Rotor Polar Moment of Inertia	Phase A - Acceleration		Phase B - At Following Conditions			Phase C - Acceleration	
	Drive Speed*	Time Max	Drive Speed*	Drive Torque Min	Time	Drive Speed* Min	Time
Lb-Ft^2	RPM	Sec	RPM	Lb-Ft	Sec	RPM.	Sec
690	1200	3	1200	290	10	2350	22

*Pad "A"

NOTE: The above values do not allow for flywheel windage losses, test stand drag, or inertia of the ATS/AD. In order to determine compliance with this specification, the test equipment shall be calibrated and specific test results corrected accordingly. If the acceleration specified for phase C is not developed because of test stand drag torque, the test stand inertia load may be reduced or external power may be applied to assist acceleration to cutout speed. Time for end of phase C to cutout shall not exceed 15 seconds.

4.5.9 Rated Torque Calibration:
Pad "A" Drive Speed (RPM), Min: 1200
Airflow, Max (Lb per Min): 110

3.5.10
4.5.10 Overrunning: The ATS/AD shall be overrun at a pad "A" drive speed of 7250 RPM for a total of 20 hours. Testing need not be continuous, but test periods shall not be less than 5 hours.

4.5.11.1 Low Temperature Operation:
Starter: The inlet air and the stabilized ATS/AD temperatures for the low temperature test shall be as follows:

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REQUIREMENTS (Continued)

Applicable paragraphs
of MIL-S-22518A(AS)

Inlet Air Total Pressure In. Hg Abs ±0.5	Inlet Air Total Temperature °F ±10	ATS /AD Temperature °F (Maximum)
140	260	-65

After the ATS/AD temperature has been stabilized for a minimum of 8 hours at -65°F, three consecutive cycles at -65°F ambient air temperature shall be run, Each cycle shall consist of the following phases:

Phase A: The ATS/AD shall accelerate the flywheel having a polar moment of inertia of 690 $lb\text{-}ft^2$ for a period of 8 seconds from rest.

Phase B: After a period of 8 seconds from rest, a torque load shall be applied to maintain the pad "A" drive speed at a constant value (±100 RPM) for a period of 10 seconds.

Phase C: At the end of phase B the torque load shall be removed and the ATS/AD allowed to accelerate the flywheel to cutout speed or for an additional period of 12 seconds, whichever, occurs first. Upon completion of the 12-second period or phase C if cutout speed has not been obtained, the ATS/AD shall be motored by an external source to a speed that will cause the automatic shutoff controls to operate.

Accessory Drive: After the ATS/AD temperature has been stabilized for a minimum of 8 hours at -65°, the ATS/AD shall be operated at a -65°F ambient air temperature at 7000 RPM input at pad "A" for a period of 10 hours with a 10 HP continuous load applied at pad "B" A 32 HP load shall be applied for the last 5 minutes of the 10-hour cycle.

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REQUIREMENTS (Continued)

Applicable paragraphs
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4.5.11.2 High Temperature Operation:

Starter: The inlet air pressure and temperature and the stabilized ATS/AD temperatures for the high temperature test shall be as follows:

Inlet Air Total Pressure In. Hg Abs ±0.5	Inlet Air Total Temperature °F ±10	ATS /AD Temperature °F (Minimum)
90	445	160

After the ATS/AD temperature has been stabilized for a minimum of 8 hours at +160°F, three consecutive cycles at a +160°F ambient air temperature shall be run. Each cycle shall be as specified for phases A, B, and C of 4.5.11.1.

Accessory Drive: After a minimum 8-hour soak period at an ambient air temperature of +250°F, the ATS/AD shall be operated with ambient air at 250 F minimum at 7000 RPM speed at pad "A" drive for a period of 30 minutes. A 10 HP continuous load shall be applied at pad "B." except for 2.5 minutes when a load of 32 HP shall be applied. This 30-minute cycle of operation shall be repeated until 10 hours of operating time are accumulated.

4.5.12 Consecutive Cycle Operation: No. of cycles: 5

4.5.14 Vibration: All vibration testing shall be conducted with the specified accessory or a dummy, having a weight of 17.4 pounds and-a center of gravity 4 inches from the-mounting face, mounted on the accessory pad (pad "B") . Mounting shall be dynamically similar to that encountered in service use. Vibration tests shall be conducted in accordance with MIL-STD-810, Method 514, Procedure I, Curve F (Parts 1, 2, and 3). In addition, the ATS/AD shall be capable of withstanding a loading of 10g bending moment applied at the ATS/AD e.g. in any direction without structural damage.

3.5.6
4.5.18 Pad "A" Drive Shear Section Strength: Torque (Lb-Ft): 1000 ±50

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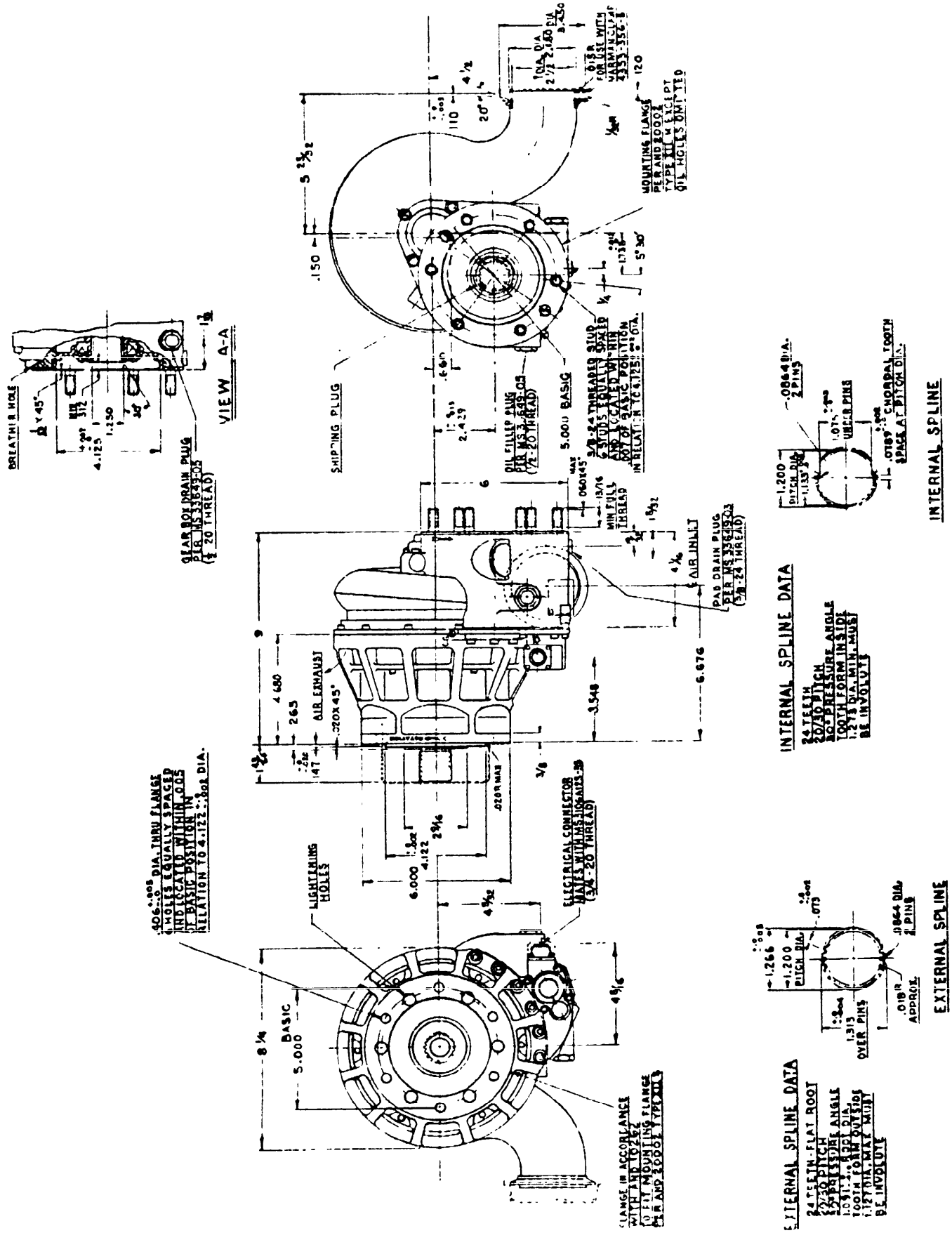


FIGURE 1. Starter, Air Turbine, Model ATS/AD-1.

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TEST SCHEDULE FOR ATS/AD-1 OPERATING AS AN ACCESSORY DRIVE

ALTITUDE (FT)	TIME (MIN)	PAD "E" DRIVE SPEED (RPM)	HYDRAULIC PUMP DISCHARGE PRESSURE (PSI)	HYDRAULIC PUMP FLOW (GPM)	ATS/AD AMBIENT AIR TEMPERATURE °F	CONDITION
Sea Level	7.5	2980	2950	13	200	Idle
Sea Level	7.5	2980	2950	0	200	Idle
SL to 45,000	2.5	4230	2950	13	200	Military
SL to 45,000	2.5	4230	2950	0	200	Military
45,000	1	4230	2950	18	200	Military
45,000	1	4230	2950	6	200	Military
45,000	5	4230	2950	0	200	Military
45,000	1	4230	2950	18	200	Military
45,000	1	4230	2950	6	200	Military
45,000	6	4230	2950	0	200	Military
45,000	1	4100	2950	18	170	*Normal
45,000	1	4100	2950	6	170	*Normal
45,000	5	4100	2950	0	170	*Normal
45 to 70,000	1	4260	2950	6	200	Afterburner
45 to 70,000	4	4260	2950	0	200	Afterburner
70,000	1	4260	2950	6	200	Afterburner
70,000	4	4260	2950	0	200	Afterburner
70 to 45,000	1	4230	2950	18	200	Military
70 to 45,000	1	4230	2950	0	200	Military
70 to 45,000	1	4230	2950	18	200	Military
70 to 45,000	1	4230	2950	0	200	Military
70 to 45,000	1	4230	2950	18	200	Military
45,000	1	4230	2950	18	170	**Normal
45 to 5000	1	4230	2950	0	170	Normal
45 to 5000	1	2980	2950	13	200	Idle
45 to 5000	4	2980	2950	0	200	Idle
45 to 5000	1	2980	2950	13	200	Idle
45 to 5000	4	2980	2950	0	200	Idle
5000	1	3680	2950	13	200	80%
5000	4	3680	2950	0	200	80%
5000	1	3680	2950	13	200	80%
5000	4	3680	2950	0	200	80%
5000	1	4230	2950	18	200	Military
5000	1	4230	2950	13	225	Military
5000	1	4230	2950	6	225	Military
5000	2	4230	2950	0	225	Military
5000	1	3600	2950	18	200	80%
5000	3	3600	2950	13	200	80%
5000	1	3600	2950	0	200	80%

Note: *Repeat for total of four cycles with zero load of last cycle for 7 minutes.

**Continue this run for 30 minutes with 1-minute cycles at 18 GPM and momentarily back to zero flow between each 1-minute loading.

FIGURE 2. Test schedule for ATS/AD-1.

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